

CET231-CyberSecurity

Fall 2024 - Lab #6

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Topics Covered:

- **Decryption**



Description:

In this lab, you will learn the basics of Decryption, algorithm description and its implementation in Python.

1. Decryption with python- Caesar Cipher

To decrypt the message, we simply need to shift each letter back by the same number of positions. In this case, you would shift each letter in "KHOOR" back by 3 positions to get the original message, "HELLO".

$$E_n(x) = (x+n) \bmod 26$$

(Encryption Phase with shift n)

$$D_n(x) = (x-n) \bmod 26$$

(Decryption Phase with shift n)

As x donates the number of the letters, as shown in the table below

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25



Example

Consider an example where the statement " GIEWIV GMTLIV HIQS " and with Key = 4 to be implemented using Caesar Cipher.

2. Cryptography with Symmetric and Asymmetric Cryptography

Symmetric Cryptography In this type, the encryption and decryption process use the same key. It is also called secret key cryptography.

Drawback

The major drawback of symmetric cryptography is that if the key is leaked to the intruder, the message can be easily changed, and this is considered as a risk factor.

○ Advanced Encryption Standard (AES)

AES is a symmetric encryption algorithm and a block cipher. Block cipher means that AES splits a message into smaller blocks and encrypts those blocks to convert the plaintext message to an unintelligible form called ciphertext.

- **The process occurs in stages:**

1. Key expansion

2. Initial round

3. Each round of AES encryption consists of four transformations:

1. Sub Bytes
2. Shift Rows
3. Mix Columns
4. Add Round Key

4. Finalization



5. Output

Decryption is performed with the reverse operations in the opposite order. The key schedule is used directly (not reversed) to derive the round keys for decryption.

Example

Write a Python program to implement AES algorithm (Encryption and Decryption).

